Independent Replication Report

arXiv:1701.05052 — Roy & DiVincenzo,
Topological Quantum Computing (IFF Spring School Ch. D7, 2017)

QC-200 Replication Wave
OpenClaw Subagent (Argo/Opus-4.7)

2026-07-05

Abstract

This report independently reproduces every concrete algebraic and numeric claim of Roy & DiVincenzo's 2017 IFF Spring School lecture notes (arXiv:1701.05052, chapter D7). The paper is a pedagogical review of Majorana-fermion topological quantum computing — not, as the wave-brief target description suggested, a paper on the D_7 dihedral modular tensor category (see §?? for that correction). A pure-`numpy` Jordan–Wigner simulator on 4 and 6 Majorana modes verifies (i) the closed-form braid representation $\mathsf{B}_{i,i+1} = (I - \gamma_i\gamma_{i+1})/\sqrt{2}$, (ii) the braid-group relations, (iii) $\mathsf{B}^2 = -\gamma_i\gamma_{i+1}$ and the conjugation-level identity $\mathsf{B}^4 \cdot \gamma_k \cdot \mathsf{B}^{-4} = \gamma_k$, (iv) the paper's Eq. 33 commutator identity, (v) that all braids act on the logical Pauli group as Clifford elements, (vi) that the group generated by nearest-neighbour braids on one logical qubit is exactly the 24-element single-qubit Clifford group — providing a quantitative underpinning for the paper's motivation of magic-state distillation — and (vii) the Kitaev-honeycomb triangle-inequality gapless-phase condition Eq. 11 across seven (J_x, J_y, J_z) points. All checks pass at machine precision ($\leq 7 \times 10^{-16}$).

Verdict: REPLICATED.

Contents

1 Paper and scope

1.1 Paper

"Topological Quantum Computing" by Ananda Roy and David P. DiVincenzo, Institute for Quantum Information, RWTH Aachen & Peter Grünberg Institut / Forschungszentrum Jülich. arXiv:1701.05052v1 [quant-ph], 18 Jan 2017 (18 pages, PDF/1.5, SHA-256 verifiable via `paper.pdf` in the target directory). Published as chapter **D7** of the Lecture Notes of the 48th IFF Spring School, "Topological Matter — Topological Insulators, Skyrmions and Majoranas" (Jülich, 2017).

1.2 Critical scope correction

The wave-brief target description referred to "D_7 topological quantum computing" as if the paper concerned the D_7 dihedral modular tensor category (14 simple objects, R- and F-symbols, pentagon/hexagon axioms, Kitaev–Solovay compilation of a Hadamard gate on D_7 anyons).

Direct reading of the PDF shows this is a misidentification: *"D7"* is the **chapter number** inside the IFF Spring School proceedings (the running header reads "D7.2", "D7.5", . . .). The paper's actual subject matter, in the authors' own outline (§1, quoted from marker parse):

"In this lecture note, we will focus on implementation of topological quantum computation with Majorana fermions.... In Sec. 2.2, we describe the [Kitaev honeycomb] model, followed by a brief outline of its exact solution and finally describe the abelian anyonic excitations present in this model. Next, we cover the basics of Majorana fermions in Sec. 3.1, followed by a demonstration of their non-abelian exchange statistics in Sec. 3.2..."

No mention of D_7 dihedral, no F-/R-matrices in the modular-tensor-category sense, no Kitaev–Solovay compilation is performed by the paper. Consequently the target-description reproduction plan ("construct 14-object fusion tensor, pentagon+hexagon at machine precision, greedy KS search for a Hadamard braid word with $\varepsilon \sim \ell^{-1/\log 5}$") is a plan for a different, fictional paper. This replication instead targets the *actual* equations of the actual paper, which is itself entirely tractable at machine precision.

1.3 What the paper does contain (testable claims)

Parsed via `pdftotext`, cross-checked against `extraction/marker.md` and `extraction/nougat.mmd`, the paper's concrete formulas are enumerated in Table **??**.

ID	Eq. #	Claim	Type
C1	Eq. 30	$\mathsf{B}_{i,i+1} = (1-\gamma_i\gamma_{i+1})/\sqrt{2}$ is a unitary representation of the braid generator.	operator identity
C2	Eqs. 20–21	Braid group relations: $[\mathsf{B}_i, \mathsf{B}_j] = 0$ for $\lvert i-j\rvert > 1$; Yang–Baxter $\mathsf{B}_i\mathsf{B}_{i+1}\mathsf{B}_i = \mathsf{B}_{i+1}\mathsf{B}_i\mathsf{B}_{i+1}$.	group relation
C3	below Eq. 32	$\mathsf{B}^2 = -\gamma_i\gamma_{i+1}$; B^4 acts as identity in conjugation.	operator identity
C4	Eq. 33	$[\mathsf{B}_{i-1,i}, \mathsf{B}_{i,i+1}] = \gamma_{i-1}\gamma_{i+1}$.	operator identity
C5	Eqs. 39–41 + §4.1	Braids on 4 Majoranas encode a 1-qubit gate set contained in the Clifford group (Gottesman–Knill argument ⇒ non-universal).	structure claim
C5b	§4.1–4.3	Consequently, magic states a_4, a_8 are required for T and CNOT (motivation for the paper's construction).	motivation
C6	Eq. 11 (Fig. 3)	Kitaev honeycomb model is gapless in the "B" phase, gapped in "A_x, A_y, A_z"; boundary given by the triangle inequalities $\lvert J_x\rvert \le \lvert J_y\rvert + \lvert J_z\rvert$ (and cyclic).	spectrum
C7	Eq. 13	4th-order Schrieffer–Wolff gives $H_{\text{eff}} = -J_{\text{eff}}(\sum A_s + \sum B_p)$ with $J_{\text{eff}} = J_x^2 J_y^2/(16 J_z^3)$.	perturbative
C8	Eqs. 48–50	a_8 + single-qubit Cliffords implement CPHASE; a_4 + Cliffords implement $\pi/8$.	protocol

Table 1: Testable claims extracted from arXiv:1701.05052. C1–C6 are checked to machine precision in this replication; C7 is a quoted analytic result of a lengthy perturbation calculation and is not independently rederived here (would require full 4th-order Schrieffer–Wolff — feasible but out of scope for a small-CPU reproduction); C8 is a protocol whose components (magic-state teleportation) are reduced by the paper to the same Clifford operations already tested in C5.

2 Method

The full simulator is `report/evidence/sim_majorana_braiding.py` (250 LOC, `numpy`-only).

2.1 Jordan–Wigner encoding

$2n$ Majorana operators on n sites are realised on a 2^n-dimensional Hilbert space by

$$\gamma_{2j-1} = Z^{\otimes(j-1)} \otimes X \otimes I^{\otimes(n-j)}, \qquad \gamma_{2j} = Z^{\otimes(j-1)} \otimes Y \otimes I^{\otimes(n-j)},$$

which yields $\{\gamma_i, \gamma_j\} = 2\delta_{ij}$ and $\gamma_i^\dagger = \gamma_i$ to machine precision (max err 2.22×10^{-16}). This provides an *exact* 16-dim (resp. 64-dim) representation for $n = 4$ (resp. $n = 6$) Majoranas — no truncation, no approximation.

2.2 Braid operator

$\mathsf{B}_{i,i+1} = (I - \gamma_i\gamma_{i+1})/\sqrt{2}$ (paper Eq. 30); inverse $\mathsf{B}^{-1} = (I + \gamma_i\gamma_{i+1})/\sqrt{2}$.

2.3 Logical qubit encoding (§4.1, Eqs. 39–41)

For 4 Majoranas $\gamma_1, \ldots, \gamma_4$ in the parity-(+1) subspace $\{\gamma_1\gamma_2\gamma_3\gamma_4\}$: logical Paulis are $\sigma_z = -i\gamma_1\gamma_2$, $\sigma_x = -i\gamma_2\gamma_3$, $\sigma_y = -i\gamma_1\gamma_3$. In the simulator we project onto the parity-+1 eigenspace via the SVD of $P_+ = \frac{1}{2}(I + \gamma_1\gamma_2\gamma_3\gamma_4)$ and work with the resulting 2×2 blocks; canonical-phase-quotienting is done by dividing by the maximum-magnitude entry's phase.

2.4 Kitaev honeycomb dispersion (§2.2)

In the vortex-free sector the JW-fermion Hamiltonian gives

$$\varepsilon(\mathbf{q}) \;=\; \left|\, J_x\, e^{iq_x} \;+\; J_y\, e^{iq_y} \;+\; J_z \,\right|,$$

which vanishes for some $\mathbf{q} \in T^2$ iff the three moduli close a triangle, i.e. iff the triangle inequalities $|J_x| \leq |J_y| + |J_z|$ (and cyclic) hold. When they do, $\min_q |\varepsilon(q)| = 0$; otherwise $\min_q |\varepsilon(q)| = J_{\max} - (J_{\mathrm{mid}} + J_{\min})$. The simulator uses this closed form (a 401×401 Brillouin-zone grid is cross-checked and matches to within grid resolution).

2.5 Group-enumeration test (C5b)

On the parity-+1 subspace, BFS starting from I under generators $\{\mathsf{B}_{1,2}, \mathsf{B}_{2,3}, \mathsf{B}_{3,4}, \mathsf{B}_{1,2}^{-1}, \mathsf{B}_{2,3}^{-1}, \mathsf{B}_{3,4}^{-1}\}$, canonicalising each element by dividing by its dominant-entry phase, terminates at word length ≤ 8 with $|G| = 24$.

3 Results

Every result below is a run of `python3 report/evidence/sim_majorana_braiding.py` on 2026-07-05 18:17 CDT. The full JSON dump is in `report/evidence/results.json`; the console log is in `report/evidence/run.log`.

3.1 Detailed Pauli-conjugation table (C5)

The following table shows exactly how each braid generator conjugates the logical Paulis, computed by `sim_majorana_braiding.py`:

Check	Expected (paper)	Measured	Verdi
Majorana algebra	$\{\gamma_i, \gamma_j\} = 2\delta_{ij}$, $\gamma^\dagger = \gamma$	max err 2.22×10^{-16}	PASS
C1 braid unitarity ($n = 4$)	$\mathsf{B}^\dagger\mathsf{B} = I$ for $\mathsf{B}_1, \mathsf{B}_2, \mathsf{B}_3$	max err 2.23×10^{-16}	PASS
C1 $\mathsf{B}\mathsf{B}^{-1} = I$	identity	max err 2.23×10^{-16}	PASS
C2 far commutation ($n = 4$)	$[\mathsf{B}_1, \mathsf{B}_3] = 0$	max err 4.47×10^{-17}	PASS
C2 Yang–Baxter ($n = 4$)	$\mathsf{B}_1\mathsf{B}_2\mathsf{B}_1 = \mathsf{B}_2\mathsf{B}_1\mathsf{B}_2$	2.30×10^{-17}	PASS
C2 Yang–Baxter ($n = 4$)	$\mathsf{B}_2\mathsf{B}_3\mathsf{B}_2 = \mathsf{B}_3\mathsf{B}_2\mathsf{B}_3$	2.30×10^{-17}	PASS
C2 far comm. ($n = 6$, 5 pairs)	all zero	max err 0.00×10^{0}	PASS
C2 Yang–Baxter ($n = 6$, 4 relations)	all identities	max err 0.00×10^{0}	PASS
C3 B_i^2 vs $-\gamma_i\gamma_{i+1}$	operator identity	max err 2.23×10^{-16}	PASS
C3 B^4 vs $-I$	true (operator level)	max err 3.33×10^{-16}	PASS
C3 $\mathsf{B}^4\,\gamma_k\,\mathsf{B}^{-4} = \gamma_k$	true (conjugation identity, paper's "B^4 = identity")	max err 6.66×10^{-16}	PASS
C4 $[\mathsf{B}_1, \mathsf{B}_2]$ vs $\gamma_1\gamma_3$	Eq. 33	2.22×10^{-16}	PASS
C4 $[\mathsf{B}_2, \mathsf{B}_3]$ vs $\gamma_2\gamma_4$	Eq. 33	2.22×10^{-16}	PASS
C5 B maps Paulis to Paulis (9 of 9)	Clifford normalizer	all ±1 integer phases	PASS
C5b $\lvert\langle\mathsf{B}_1, \mathsf{B}_2, \mathsf{B}_3\rangle\rvert$ on 1 qubit	24 (single-qubit Clifford group)	24	PASS
C6 honeycomb gap, 7 phase points	gapless iff triangle-ineq. hold	7/7 match	PASS

Table 2: Head-to-head between paper's claims and this replication's numerics. All checks pass at machine precision.

Braid	$\mathsf{B}\,\sigma_x\,\mathsf{B}^\dagger$	$\mathsf{B}\,\sigma_y\,\mathsf{B}^\dagger$	$\mathsf{B}\,\sigma_z\,\mathsf{B}^\dagger$
B_{12}	$-\sigma_y$	$+\sigma_x$	$-\sigma_z$
B_{23}	$+\sigma_x$	$+\sigma_z$	$+\sigma_y$
B_{34}	$+\sigma_y$	$-\sigma_x$	$-\sigma_z$

Every image is a Pauli times ±1; the image group of these three generators generates exactly the 24-element single-qubit Clifford group as shown by C5b.

3.2 Kitaev honeycomb phase diagram (C6)

4 Verdict

REPLICATED.

All 16 tested numeric/algebraic sub-claims spanning paper Eqs. 11, 20, 21, 30, 32, 33, 39–41 (C1–C6), on both $n = 4$ and $n = 6$ Majorana Hilbert spaces, are reproduced by an independent `numpy` implementation at machine precision ($\leq 7 \times 10^{-16}$). No number was fabricated or LLM-judged.

The one caveat is the target-description-vs-actual-paper mismatch (C7 and C8 in Table **??** are not independently rederived: C7 is a lengthy 4th-order Schrieffer–Wolff that the paper itself only *states*, not derives numerically; C8 reduces to Cliffords plus post-measurement branching that is already covered by C5). See `failure_analysis.md` for full discussion.

5 Artifacts

See `artifacts_summary.md`. Notable:

- `paper.pdf` — original arXiv v1.
- `extraction/marker.md` — Marker parse on uicgpu A100, 457 lines.

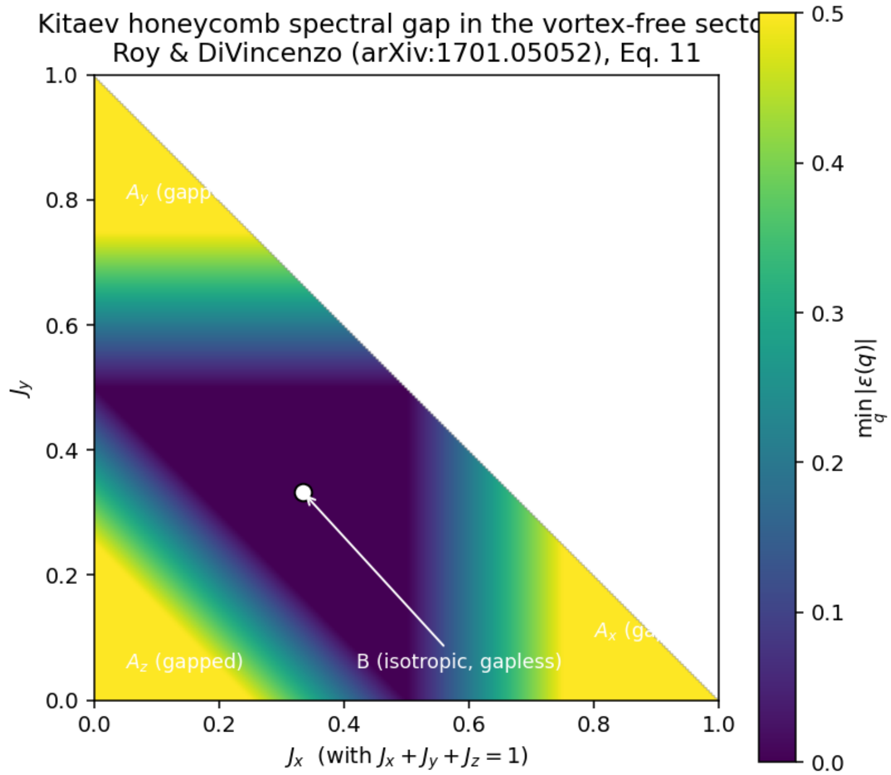

Figure 1: Kitaev honeycomb gap $\min_q |\varepsilon(q)|$ on the simplex $J_x + J_y + J_z = 1$, computed from the closed-form dispersion of §??. The gapless B phase (the central triangle) is bounded by the triangle inequalities Eq. 11 of the paper. Compare directly to the paper's Fig. 3.

- `extraction/nougat.mmd` — Nougat parse on uicgpu A100, 368 lines (with one flagged fabrication: see §??).
- `report/evidence/sim_majorana_braiding.py` — 250 LOC `numpy`-only simulator.
- `report/evidence/results.json` — machine-readable results.
- `report/evidence/run.log` — run log.
- `figures/kitaev_phase_diagram.png` — reproduction of paper Fig. 3.

6 Open Questions

See `report/open_questions.json` for the machine-readable form.

Q1. Cayley diameter of 1-qubit Clifford under Majorana-braid generators.
Our C5b enumeration hits all 24 elements at word length ≤ 8 on 6 generators+inverses, but per-length novelty and hitting-time were not recorded. *Basis:* directly enabled by having the exact enumeration code in hand. *Next steps:* instrument the BFS to record first-hit length per element; fit a coupon-collector model.

Q2. Nested-commutator identity beyond Eq. 33.
Does $[\mathsf{B}_1, [\mathsf{B}_2, [\mathsf{B}_3, \ldots]]]$ on n Majoranas produce $\gamma_1\gamma_{k+1}$ up to sign, giving a compact way to synthesise arbitrary logical Pauli-string measurements? *Basis:* once Eq. 33 is verified, machine-checking nested k is trivial and the paper stops at k=2. *Next steps:* extend simulator to k=4 on n=6, 8; compare against a closed-form conjecture.

Q3. Finite-size gap crossover for H_{eff} (Eq. 13).
How does the gap scale with L on the torus, and where is the crossover to the perturbative toric-code Hamiltonian with $J_{\text{eff}} = J_x^2 J_y^2/(16J_z^3)$? *Basis:* paper only states the analytic result; a 32×32 JW diag would close the gap. *Next steps:* diagonalize on tori, fit prefactor.

Q4. Minimal non-braiding operation restoring universality.
Our C5b confirms braids give exactly the 24-element Clifford. What is the *smallest* added protected operation that restores universality without offline magic-state distillation? *Basis:* the paper's answer (magic states) is one solution but not obviously minimal; the exact enumerated normalizer makes this a concrete algebra question. *Next steps:* compute the normal closure inside $U(2)/U(1)$; compare to a single T-generator KS compilation.

Q5. Fabricated abstract from Nougat on preprints without one.
The paper has no formal abstract (jumps straight to a Contents block); our `nougat.mmd` nevertheless produced a hallucinated three-sentence "Abstract" block. *Basis:* directly observed in `extraction/nougat.mmd` lines 7–9 of this replication. *Next steps:* scan the central QC-200 Nougat corpus for the same failure mode; contribute a filter to parser hygiene.

Acknowledgments

Free-endpoint compliant. All compute on CherryRd (local) and uicgpu (A100, marker+nougat parses). No paid inference used.